

GroupTrack

User Guide — Version 2.4

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GroupTrack is a custom Android app for coordinating multi-vehicle group rides without cellular or internet connectivity. It uses Meshtastic mesh radio hardware to share GPS positions between all vehicles in real time. No phone signal required. No server. No manual radio configuration.

1. USER ROLES

ROLE	DESCRIPTION
ORGANIZER	Creates rides, configures radios, distributes ride configs. Has access to developer settings panel via long press on GroupTrack header.
RIDER	Receives ride config, imports it, connects radio, applies config automatically. Standard user role.

2. SPLASH SCREEN

When GroupTrack launches for the first time or after restart, a branded splash screen displays while the app initializes and waits for mesh nodes. The splash shows the GroupTrack logo, UTV desert background, developer credit, and website.

The splash automatically dismisses when the first mesh node is detected or after a brief initialization period.

3. MAIN MAP SCREEN

The main screen shows a live map with all convoy vehicles plotted as labeled markers. This is the primary operational screen during a ride.

ELEMENT	DESCRIPTION
GroupTrack Header	Tap to open the GroupTrack submenu. Long press to access developer settings (organizer only, password required).
RECORD Button	Top-left red dot. Tap to start GPS track recording. Tap again to access pause/stop options.
Map Settings Panel	Top-right panel showing MAP + current source (e.g. MAP SAT) with ONLINE/LOCAL indicator. Tap to expand layer, zoom, search, and offline controls.

Node Markers	Each vehicle shown as a labeled circle. Your cart is larger. Tap any marker to open NODE detail panel.
HUD Overlay	Transparent info panel over the map. Shows convoy or cart stats depending on mode selected.
Button Bar	Fixed bar at bottom: GROUP MY CART HIDE GEAR
Download Progress Chip	Appears above button bar during tile download. Shows tile count, progress, and time remaining.

MAP TYPES: SAT = Esri satellite imagery. HYB = Google satellite + roads hybrid. TOPO = Esri topographic with terrain. RD = Carto road map with points of interest.

4. BUTTON BAR

BUTTON	DESCRIPTION
GROUP	Shows convoy-wide HUD: all carts, span, lead, tail, active/lost count, channel utilization, GPS interval slider.
MY CART	Shows your vehicle's individual stats: speed, heading, battery, altitude, distance to lead/tail.
HIDE	Hides the HUD — map-only view.
GEAR (⚙)	Opens map settings: map type, zoom controls, search, offline maps, download region.

5. GROUP HUD PANEL

The GROUP HUD displays convoy-wide statistics as a transparent overlay on the left side of the map.

FIELD	DESCRIPTION
Ride Title	Name of the active ride.
UNITS	Total number of cart nodes on the mesh.
ACTIVE	Vehicles heard recently.
LOST	Vehicles that have gone silent.
LEAD	Callsign of the frontmost vehicle (green). Determined automatically by position.
TAIL	Callsign of the rearmost vehicle (orange). Determined automatically.
SPAN	Distance in miles from LEAD to TAIL.

CH%	Channel utilization percentage from radio telemetry. Green <25%, Amber 25-40%, Red >40%.
INT Slider	GPS broadcast interval slider (2-8 seconds). Vertical slider on left side of HUD.

6. MY CART HUD PANEL

FIELD	DESCRIPTION
Callsign	Your convoy callsign (e.g. INDIA-7 or HOTEL-10 for operator).
SPD	Current speed in mph.
HDG	Heading in degrees (0-359).
BAT	Radio battery percentage.
ALT	Altitude in feet.
AHEAD / BEHIND	Distance to nearest vehicles ahead and behind you in the convoy.
TO LEAD / TO TAIL	Distance to front and back of the convoy.

7. NODE HUD — INDIVIDUAL CART DETAIL

Tap any cart marker on the map to open the NODE detail panel for that vehicle.

FIELD	DESCRIPTION
Cart Name	Convoy callsign of the selected cart.
Status	ACTIVE, SIGNAL DROP, or LOST.
SPD / HDG / BAT / ALT	Speed, heading, battery, altitude of the selected cart.
SNR	Signal quality of the last received radio packet.
Position in convoy	Order position, e.g. 3 of 8.
Last seen	Time since last position packet.
RETURN	Returns to full convoy view.
SET AS LEAD	Designates this cart as the LEAD vehicle.
REMOVE FROM RIDE	Removes cart from active ride roster.

8. MAP SETTINGS PANEL

Tap the map panel header (top right, showing MAP + source name) to expand the full settings panel. The header always shows the current tile source and ONLINE/LOCAL status.

8.1 Layer Selection

Four tile source buttons: SAT | HYB | TOPO | RD. Tap to switch. The active source is highlighted blue. Switching to an online source sets the indicator to ONLINE. Switching to offline mode serves tiles from local cache.

8.2 Download Zoom

Slider controlling the maximum zoom level for tile downloads. Range z16-z18. Higher zoom = more detail but more tiles and larger storage.

8.3 Fly Radius

Slider controlling how close the map zooms when flying to a search result. Range 1-10 miles. 10 miles gives a regional view for selecting a download zone. 1 mile gives a tight trail-level view.

8.4 Online / Offline Switch

Toggle between ONLINE (streaming tiles from internet) and LOCAL (serving tiles from device storage). Set to LOCAL before heading into the field. Set to ONLINE when back in connectivity.

The ONLINE/LOCAL indicator in the map header always reflects the current switch state: green = ONLINE, blue = LOCAL.

8.5 Location Search

Search for a named area before downloading map tiles. Type a location name (city, park, trail system, or region) and tap FIND AREA.

STEP	DESCRIPTION
1. Type location	Enter city, park name, or trail area. Examples: Zion National Park UT, Gooseberry Mesa, Leeds UT.
2. Tap FIND AREA	Geocoder searches for matching locations. Returns up to 5 results.
3. Select result	If multiple results, tap the correct one from the list. Single results fly directly.
4. Map flies to center	Map zooms to the search result center point. A blue pin marks the location with coordinates. Blue crosshair lines show the lat/lon reference.
5. Zoom and explore	Use pinch-to-zoom to explore trails, trailheads, and staging areas. The center pin stays visible as reference.

6. Tap DOWNLOAD REGION	Activates the rectangle draw tool. Draw your download area. The center pin and crosshairs remain visible as a reference while you draw. They clear when you collapse the map settings panel.
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If the geocoder cannot find your location, try adding the state abbreviation (e.g. 'Sand Hollow UT'). For precise locations, raw coordinates are supported: 37.44, -113.25.

8.6 Download Region

Tap DOWNLOAD REGION to activate the rectangle draw tool on the map. Draw a box over the area you want to cache for offline use. When you release, a size estimation dialog appears.

DIALOG ELEMENT	DESCRIPTION
Dimensions	Area width x height in miles (e.g. 3.2 mi x 2.1 mi).
Tile count	Number of tiles to download across all zoom levels.
Estimated size	Storage estimate in MB based on average tile size.
Source	Which tile layer will be downloaded (e.g. Source: SAT).
DOWNLOAD button	Starts the download. Gear panel closes. Toast notification confirms download started.
CANCEL button	Cancels without downloading.
500MB ceiling	If estimated size exceeds 500MB, the download is blocked. Reduce the selected area and try again.

8.7 Download Progress

During download, a progress chip appears above the button bar on the map screen showing tile count (X of Y tiles) and estimated time remaining. The gear panel is closed — you can continue using the map while tiles download.

A toast notification warns to keep the app open while downloading. When complete, a completion toast confirms the tile count downloaded.

Keep app open during download. Background download is not yet supported — closing the app will stop the download. Resume is automatic — tiles already on disk are skipped if you restart a download over the same area.

8.8 Show / Hide Downloaded Areas

The SHOW DOWNLOADED button in the map settings panel displays a blue overlay on the map showing all areas where offline tiles have been downloaded to the device. Use this to identify coverage gaps before heading into the field and to plan adjacent download areas without overlap.

Tapping SHOW DOWNLOADED scans all downloaded tile areas and draws them as semi-transparent blue rectangles on the map. The overlay moves and scales with the map as you pan and zoom. Tapping HIDE DOWNLOADED removes the overlay. The button toggles between these two states.

The overlay remains visible while you tap DOWNLOAD REGION and draw a new download rectangle. This lets you draw adjacent areas without gaps. Use the SHOW DOWNLOADED overlay as your guide, then draw the next rectangle to fill uncovered terrain.

NOTE: The overlay is computed at zoom level 14 for performance. Each blue rectangle represents approximately a 2.5 square mile area. For precise field-level coverage, zoom in past zoom 16 with offline mode active to confirm tile rendering directly.

9. GROUPTRACK MENU

Tap the GroupTrack header at the top of the map to open the submenu. This is the main navigation hub for ride management and radio configuration.

EVENT / RIDE Menu Items

ITEM	DESCRIPTION
Create a Ride	Create a new ride with name, date, and auto-generated encrypted channel. Organizer only.
Transfer Event/Ride Details via Email	Select a ride and send to riders as a .convoy email attachment.
Apply Ride Info to Radio	Configure the connected radio for MASTER baseline or a specific ride.
Restore Prior Config Post Ride	Select and restore a saved radio backup from the archive list.

10. CREATE A RIDE (ORGANIZER)

GroupTrack menu → EVENT / RIDE → Create a Ride

FIELD	DESCRIPTION
Ride Name	Name for the ride (e.g. HIGH DESERT RUN 2026).
Ride Date	Scheduled date. Used for sorting in the ride list.
Description	Optional notes: route, meeting point, etc.
Channel Name	Auto-generated: CONVOY-XXXX format. Can be edited. Max 11 characters.
Encryption Key	Auto-generated AES-256 key. Unique per ride.

NOTE: The channel name and encryption key are generated once. Screenshot or note them before leaving — they cannot be regenerated.

11. APPLY RIDE INFO TO RADIO

GroupTrack menu → EVENT / RIDE → Apply Ride Info to Radio

This screen configures the connected radio for use in a convoy. Two modes are available:

MODE	DESCRIPTION
MASTER	Resets radio to GroupTrack baseline: US region, LONG_FAST, 903 MHz (channel 5), 60s GPS, Convoy channel, AES-256 master key.
RIDE	Applies master baseline plus the selected ride's unique channel name and AES-256 encryption key.

Apply Process — Version 2.4 Single-Boot Flow

1. Tap PROCEED TO UPDATE — app writes full config including channel and PSK in one binary install.
2. 60-second countdown — radio reboots to apply binary config.
3. Auto-reconnect — app automatically cycles Bluetooth at 15s/12s remaining to ensure clean reconnect.
4. Auto-proceed to Verify — no user action required, app navigates directly to verification.
5. Verify screen — reads all 40 fields back from radio and shows any failures.

V2.4: Channel name, encryption key (PSK), and all 38 config fields travel in a single binary install. One reboot. No second stage required. Total process takes approximately 90 seconds.

12. VERIFY CONFIG SCREEN

After radio configuration, the Verify screen reads all 40 config fields back from the radio and compares them against expected values. Only failed fields are shown. A clean verify shows a success summary.

FIELD GROUP	FIELDS VERIFIED
Device (4)	Node Role, Managed Mode, Serial Enabled, Long Name.
LoRa (8)	Region, Preset, Bandwidth, Spread Factor, Coding Rate, Hop Limit, TX Enabled, TX Power.
Channel (4)	Channel Name, Encryption Key (32-byte AES-256), Position Precision, Mute Status.

Position (6)	Broadcast Interval, GPS Update, Smart Position, Smart Min Interval, Smart Min Distance, Fixed Position.
Display (4)	Units, Screen Timeout, Auto Brightness, Compass North Top.
Module (14)	Telemetry, MQTT, Serial, Notifications, Range Test, Store/Forward, Neighbor Info, Audio, and more.

IMPORTANT: If Channel Name or Encryption Key fail, the channel write did not complete. Run Apply Radio again. If all other fields pass but channel fails, the radio is on the wrong channel and cannot communicate with the convoy.

13. GPS TRACK RECORDING

GroupTrack can record your GPS track during a ride for post-ride analysis and route documentation.

ACTION	DESCRIPTION
Start Recording	Tap the red RECORD dot (top left of map). Recording begins immediately.
Pause Recording	Tap RECORD while recording — select PAUSE from the menu.
Stop and Save	Tap RECORD while recording — select STOP. File save dialog appears. Name your track file.
Track File Format	GPX format. Saved to device storage. Can be imported into mapping apps.

GPS recording runs independently of the mesh network. You can record your track while also viewing convoy positions on the map.

14. RADIO CONFIG REFERENCE

FIELD	VALUE
Region	US (915 MHz)
Modem Preset	LONG_FAST
Frequency	903.125 MHz (channel 5 in US915 band)
Hop Limit	3
TX Power	27 dBm
GPS Broadcast	60 seconds (adjustable 2-8s via HUD slider)
Smart Position	Enabled — fires on 100m movement

Display Units	Imperial — miles, feet, mph
Node Role	TRACKER
Position Precision	32 (full precision — exact GPS shared)
Encryption	AES-256 — unique 32-byte key per ride

15. TROUBLESHOOTING

PROBLEM	SOLUTION
No vehicles on map	Check radio is connected (green status). Verify all radios use same channel name and PSK. Run Verify Config to confirm.
Yellow lock icon on radio	Encryption not active. Run Apply Radio again — complete the full process.
Verify screen shows failures	Channel write incomplete. Run Apply Radio again from GroupTrack menu.
Radio does not reconnect	Toggle Bluetooth OFF then ON in Android notification shade. Wait 10 seconds.
Map tiles not loading	Check ONLINE/LOCAL switch in map settings. If ONLINE, check WiFi or cell data. If LOCAL, verify tiles have been downloaded for this area.
Map shows blank tiles offline	No tiles downloaded for this area or zoom level. Use SHOW DOWNLOADED in map settings to see which areas are covered. Return to connectivity, search the area, draw a download zone, and download tiles before going offline.
Download stuck at 0 tiles	Check internet connection. Ensure app has network access. Try a smaller area first.
Download blocked — too large	Area exceeds 500MB ceiling. Reduce the selected rectangle and try again.
RECORD button not tappable	Cart messages overlay may be covering the button. Wait for messages to dismiss.
App closes when tapping BT settings	Known issue. Restart Apply Radio from the menu.
Imported ride not in RIDE tab	Open GroupTrack menu — splash screen confirms import. If no splash, resend the ride file.
Location search returns no results	Add state abbreviation (e.g. 'Zion UT'). For precise locations use coordinates: 37.44, -113.25.